Hybrid Quantum–Classical Physics-Informed Neural Network for Czochralski Crystal Growth

Study report — surrogate training, ibm_fez hardware inference, and controlled ablation

Aswin

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The Industrial Problem

- Czochralski (CZ) crystal growth is the dominant method for producing single-crystal silicon ingots for semiconductor wafers.
- A seed crystal is pulled from a molten silicon bath in a rotating quartz crucible at $\sim$1685–1750 K.
- Crystal quality (defect density, oxygen concentration, resistivity uniformity) is controlled by melt hydrodynamics and thermal structure.
- Modelled with axisymmetric, incompressible Navier–Stokes equations, configured here with:
  - $Re = 15$, $Gr = 8{,}000$ (buoyancy), $Pr = 28.463$ (heat transfer)
  - Rotation-driven swirl

Five coupled output fields: u_r, u_z, u_θ, p, T

CFD solvers are accurate but slow (hours–days per simulation) — prohibitive for real-time control or parametric studies.

These dimensionless numbers are the values passed to the training code; they have not been checked against the CFD case set-up (see Appendix, "Unverified items").

The Proposed Solution

Hybrid Quantum–Classical Physics-Informed Neural Network (HQPINN) as a surrogate model:

- Takes spatial coordinates (r, z) as input
- Produces all five flow fields in milliseconds
- Trained on CFD data; Navier–Stokes residuals can be added to the loss
- Uses a 16-qubit variational quantum circuit as a feature-extraction layer
- Quantum features evaluated on real IBM hardware (inference and noise study); hardware fine-tuning was planned but **not run**

Hypothesis under test: classical expressivity + physics consistency + quantum feature representations improve the surrogate.

Why Quantum Computing?

- Quantum layer: **96 of 126,021 parameters (0.076%)**, in parallel with a 64-wide classical skip connection
- Motivation: entanglement-based correlations that are hard to reproduce classically

What the measurements say (later sections)

A controlled ablation (720 runs at 8 qubits, 120 paired comparisons per contrast, plus 12 and 16 qubits) finds **no benefit from entanglement or from training the quantum weights**, and the quantum branch is measurably worse than a matched classical branch.

This report therefore documents the infrastructure, the real-hardware noise measurements, and the defects found along the way — not a quantum advantage.

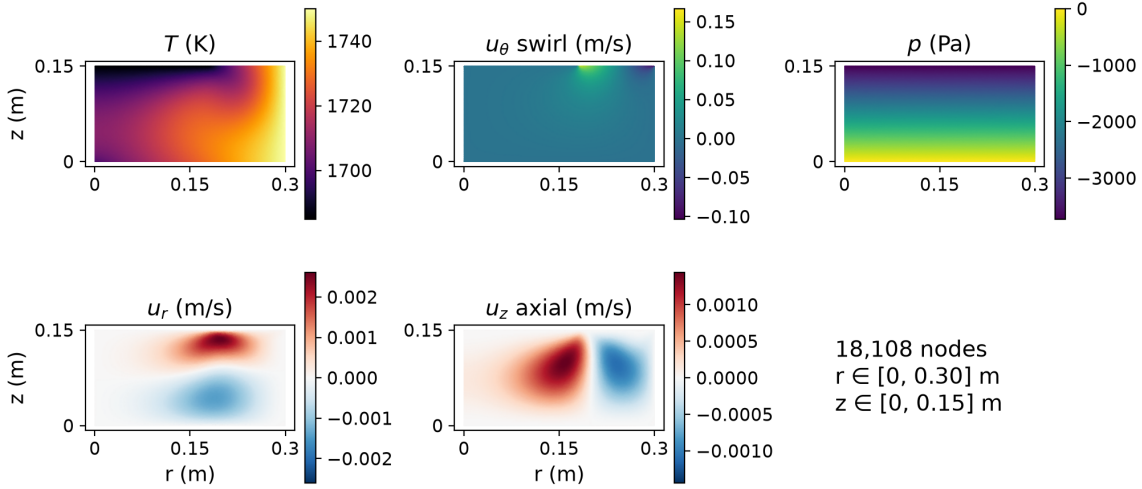
Dataset Overview

Source: COMSOL 6.2 axisymmetric simulation of the CZ melt (`cz_melt_raw.txt`). Domain: $r \in [0, 0.3\ \text{m}]$, $z \in [0, 0.15\ \text{m}]$.

Property	Value
Total data points	18,108
Spatial columns	r, z
Output columns	u_r, u_θ (swirl), u_z (axial), p, T
Temperature range	1685–1750 K
Pressure range	-3722 to 0 Pa

Train/Validation split (95/5, seed 42): 17,203 training points; 905 held out.

CFD fields in physical units



18,108 nodes
r ∈ [0, 0.30] m
z ∈ [0, 0.15] m

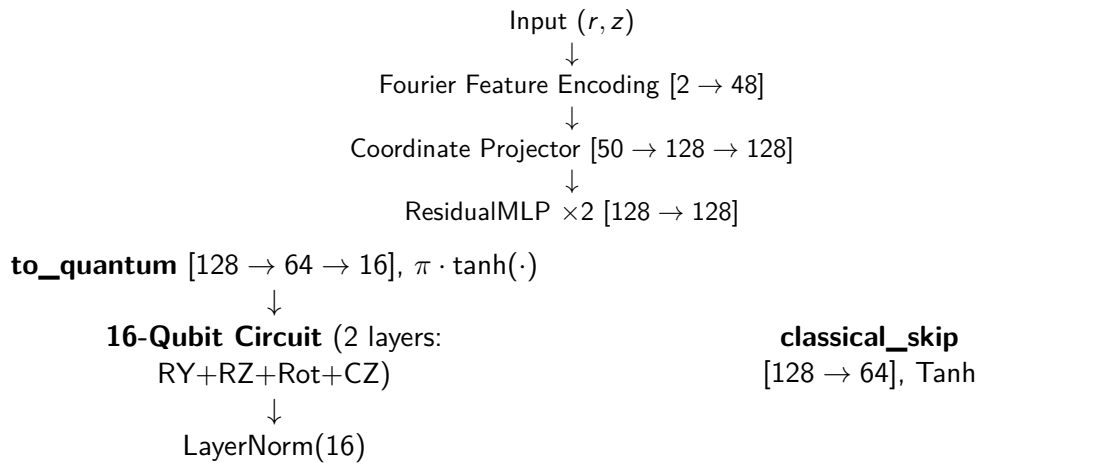
Data Normalization

Scale	Value	Derivation
Length scale	0.300 m	$\max(r_{max}, z_{max})$
Velocity scale	0.1666 m/s	max over *all three* velocity components
Pressure scale	3722.41 Pa	max absolute pressure
Temp. min / max	1685.0 / 1750.0 K	$\min(T)$, $\max(T)$
Pressure coeff. p_{coeff}	134,128.5	p_{scale}/v_{scale}^2

Shared velocity scale hides the small components

The scale is set by the swirl ($|u_\theta|_{max} = 0.1666$ m/s). The meridional components peak at only $|u_r| = 2.6$ and $|u_z| = 1.4$ mm/s, i.e. 1.6% and 0.9% of that scale, so their normalised MSE ($\sim 10^{-6}$) looks excellent while their physical-unit R^2 is only 0.87 and 0.84 (Stage 2 results).

Architecture Overview



Component Details (1/2)

Fourier Feature Encoding

- Fixed random projection $B \in \mathbb{R}^{2\times 24}$, sampled $\mathcal{N}(0, 6^2)$; in the v2 checkpoint $|B|_{max} = 19.2$
- $\text{ff}(x) = [\sin(2\pi xB), \cos(2\pi xB)]$

Classical Encoder

- Linear(50, 128) + Tanh, Linear(128, 128) + Tanh
- ResidualMLP$\times$2: $h \leftarrow \tanh(h + W_2 \cdot \tanh(W_1 \cdot h))$

To-Quantum Projection

- Linear(128, 64) + Tanh + Linear(64, 16), scaled by $\pi \cdot \tanh(\cdot)$ to $[-\pi, \pi]$

Component Details (2/2)

16-Qubit Variational Quantum Circuit: RY encoding of the 16 inputs, then 2 layers, each:

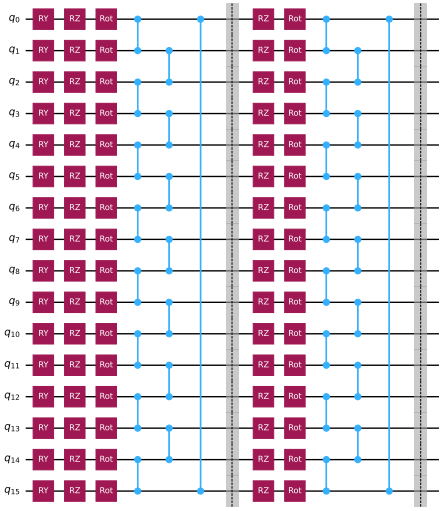
1. Data re-upload: $RZ(0.5 \cdot \theta_{(i+\text{layer}) \bmod 16})$
2. Trainable rotation: $\text{Rot}(\varphi_i, \theta_i, \omega_i)$ — 48 params/layer
3. Entangling: CZ on even pairs, odd pairs, plus a closing CZ (15–0) — 16 CZ per layer

Measurement: $\langle Z_i \rangle$, $i = 0..15$ $\Rightarrow$ **96 quantum parameters**

Classical skip connection: Linear(128,64)+Tanh bypasses the quantum layer entirely.

Hard constraints (as coded): output 0 and output 2 are multiplied by r, forcing them to vanish on the symmetry axis. In the v2 checkpoint, output 2 was fitted to the *axial* velocity (see "Defects in the v2 run"), so the constraint sits on the wrong physical field.

Logical quantum circuit (16 qubits, 2 layers)



Parameter Count

Component	Parameters
Fourier feature buffer B (frozen)	48
coord_proj (2 layers)	23,040
res1.block (2 layers)	33,024
res2.block (2 layers)	33,024
to_quantum (2 layers)	9,296
classical_skip	8,256
q_layer.weights	**96**
q_norm (scale + bias)	32
post (3 layers)	19,205
Total (incl. buffer)	**126,021**
Quantum fraction	**0.076%**

Governing Equations (as implemented)

Steady, axisymmetric, incompressible, nondimensionalised; $\rho = 1$:
Continuity:

$$\frac{\partial (r u_r)}{\partial r} + r \frac{\partial u_z}{\partial z} = 0$$

Radial momentum:

$$u_r u_{r,r} + u_z u_{r,z} - \frac{u_\theta^2}{r} + p_{coeff}\, p_r - \frac{1}{Re}\left[u_{r,rr} + \frac{u_{r,r}}{r} - \frac{u_r}{r^2} + u_{r,zz} \right] = 0$$

Axial momentum:

$$u_r u_{z,r} + u_z u_{z,z} + p_{coeff}\, p_z - \frac{1}{Re}\left[u_{z,rr} + \frac{u_{z,r}}{r} + u_{z,zz} \right] - \frac{Gr}{Re^2}\theta = 0$$

where $\theta = (T - T_{min})/(T_{max} - T_{min})$, $Re = 15$, $Pr = 28.463$, $Gr = 8000$, $p_{coeff} = 134{,}128$. **There is no gravity term** ρg; the buoyancy term acts on θ only.

Governing Equations (cont.)

Swirl (angular momentum):

$$u_r u_{\theta,r} + u_z u_{\theta,z} + \frac{u_r u_\theta}{r} - \frac{1}{Re}\left[u_{\theta,rr} + \frac{u_{\theta,r}}{r} - \frac{u_\theta}{r^2} + u_{\theta,zz}\right] = 0$$

Energy:

$$u_r \theta_r + u_z \theta_z - \frac{1}{Pr \cdot Re}\left[\theta_{rr} + \frac{\theta_r}{r} + \theta_{zz}\right] = 0$$

Implementation:

- All derivatives via `torch.autograd.grad` with `create_graph=True`
- **Quantum detachment:** q_{out} detached in the physics loss — avoids second-order gradients through the statevector
- EMA ($\beta = 0.95$) "balancing" of the 5 residual terms (found to be a no-op, see defects)

Stage 1 — Mac M4 Pipeline Check

A quick local run before committing to GPU compute (Mac Mini M4, 24 GB).

Configuration:

- 8 qubits, 1 quantum layer; PennyLane `default.qubit`, `backprop`, parameter broadcasting
- Batch size 2,048, 120 epochs (about 29 minutes, from earlier notes; no log kept)

What the script does: data-only loss, OneCycle LR, logs the training loss only. It has **no physics loss, no validation split, no per-field logging and no resume**.

What it tested: data loading, normalisation, the quantum layer under broadcasting, checkpoint saving. **No validation MSE was measured**, so no accuracy is claimed for this stage.

Compute Infrastructure — Brev H100 DGX

Resource	Specification
GPU	NVIDIA H100 80GB HBM3 (GPU 4 of 8)
Storage	/data volume
OS	Ubuntu 22.04
CUDA driver	580.105.08 (CUDA 13.0)
PyTorch	2.11.0+cu130
PennyLane	0.44.1

GPUs 0–3 ran a vLLM inference workload; GPU 4 was used for this training. Launched via `nohup` with `CUDA_VISIBLE_DEVICES=4`.

Training Configuration

Hyperparameter	Value	Note
Epochs	400	221.2 min on H100
Batch size	256	68 batches/epoch
Learning rate	2×10^{-3}	CosineAnnealing $\rightarrow 10^{-6}$
Physics warmup	epoch 50	data-only before
Physics ramp	**150 epochs**	linear $0 \rightarrow 0.05$, full weight at epoch 200
Physics weight	0.05 (max)	
Grad clip	1.0 (L2)	
EMA beta	0.95	"balancing" (no-op)
Train/val split	95/5	17,203 / 905
Checkpointing	every 50 epochs + best-val	

Phase 1: Data-Only Training (Epochs 1–50)

Epoch	Train Loss	Val (sum)	u_r	u_θ swirl	u_z axial	p / T
1	1.16e-2	4.83e-3	4.40e-5	1.69e-3	4.44e-5	1.68e-3 / 1.37e-3
10	5.92e-5	3.11e-4	3.61e-6	8.57e-5	2.46e-6	7.24e-5 / 1.47e-4
20	**5.26e-5**	**1.58e-4**	**1.93e-6**	**5.78e-5**	**1.44e-6**	**4.62e-5 / 5.07e-5**
30	2.96e-5	1.75e-4	1.20e-6	6.51e-5	1.12e-6	4.58e-5 / 6.15e-5
40	4.43e-5	3.06e-4	2.77e-6	1.79e-4	9.15e-7	6.22e-5 / 6.16e-5
50	2.07e-5	2.13e-4	1.06e-6	7.42e-5	9.05e-7	1.07e-4 / 2.91e-5

"Val (sum)" is the *sum* of the five per-field normalised MSEs (the mean at epoch 20 is 3.2×10^{-5}). Swirl and axial columns are labelled by physical meaning; the log calls them u_z and u_θ (transposed loader, see defects).

Best Result

Best total of $\mathbf{1.58 \times 10^{-4}}$ at **epoch 20**, saved as `pretrained_v2.best.pt`. Re-evaluating that checkpoint reproduces all five logged per-field values exactly.

Phase 2: Physics Loss Ramp (Epochs 50–200)

Epoch	Phys. Weight	Train Loss	Phys. Loss	Val p / T
60	3.33e-3	1.01e-1	20.05	1.05e-1 / 7.53e-2
70	6.67e-3	6.40e-2	4.24	1.06e-1 / 7.93e-2
80	1.00e-2	7.42e-2	3.79	1.06e-1 / 7.70e-2
100	1.67e-2	1.50e-1	7.19	1.06e-1 / 1.05e-1
110	2.00e-2	2.45e-1	11.13	2.22e-1 / 1.14e-1
150	3.33e-2	1.20e-1	2.57	1.05e-1 / 8.41e-2
200	5.00e-2	1.09e-1	1.37	1.06e-1 / 1.02e-1

Critical Observation

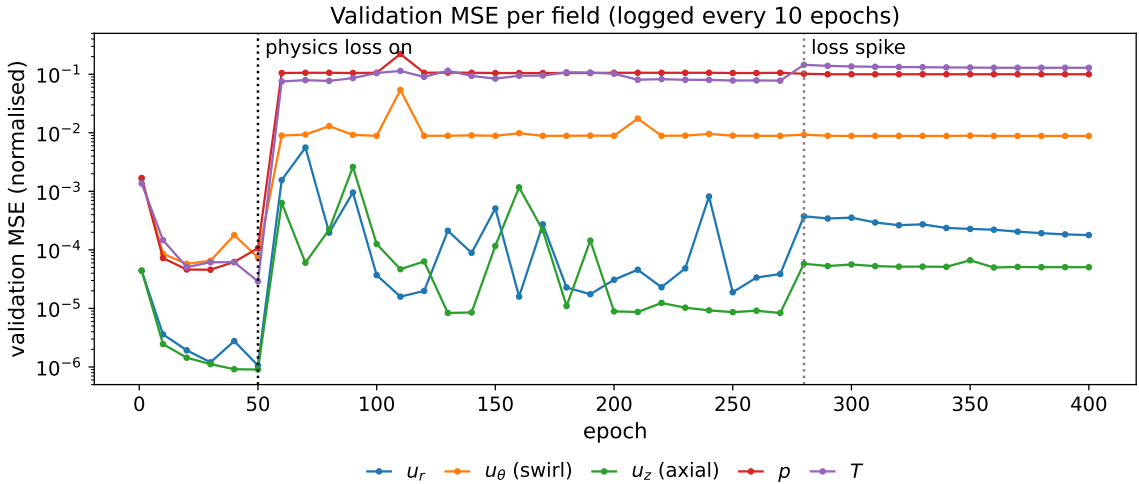
When the physics loss switched on, validation MSE for p and T jumped from $\sim 5 \times 10^{-5}$ to ~ 0.1 — **three orders of magnitude**. The weight was still only 3×10^{-3}.

Phase 3: Full Physics Weight (Epochs 200–400)

Epoch	Train Loss	Physics Loss	Val (sum)
250	8.72e-2	9.75e-1	0.193
270	6.04e-2	4.19e-1	0.192
280	**1.04e+9**	**2.16e+10**	0.256
290	**1.18e+14**	**2.15e+15**	0.248
300	5.57e-2	9.62e-2	0.246
350	5.42e-2	4.79e-2	0.239
400	4.92e-2	2.04e-3	0.238

At epochs 280–290 the total loss reached 10^9–10^{14} (cause not diagnosed). Gradient clipping kept the parameters bounded and the loss was back to normal by epoch 300. Final validation stayed at $\sim$0.238, **1,500$\times$ worse than the best checkpoint**.

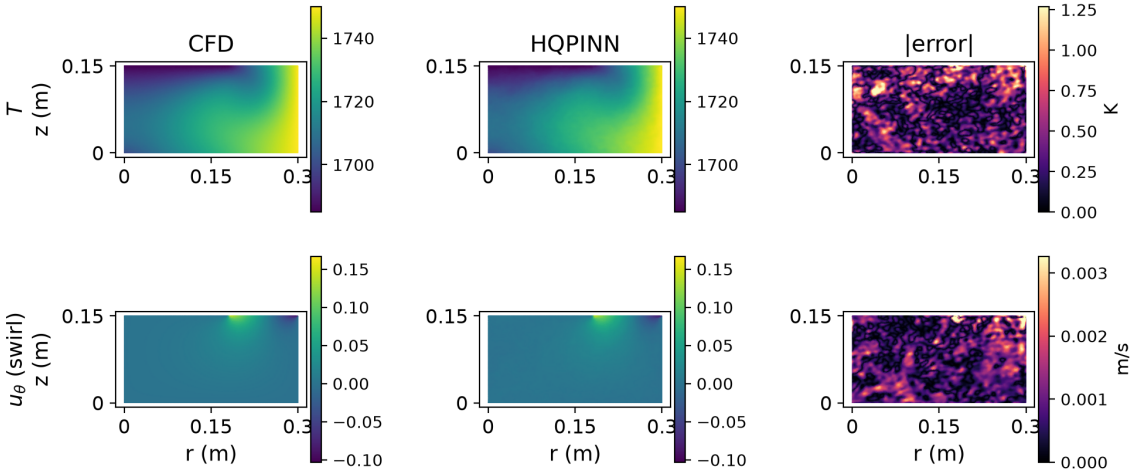
Per-field validation MSE through training



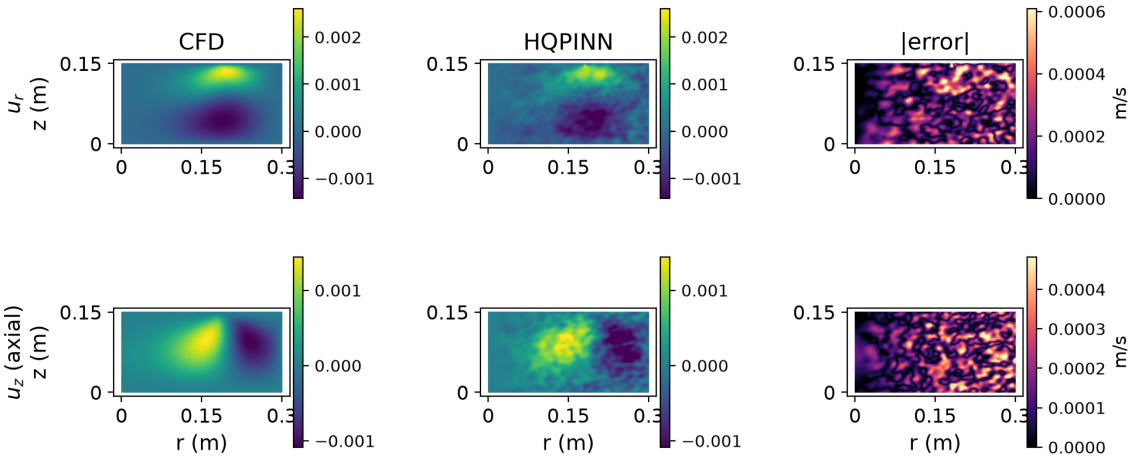
Stage 2 Summary

Metric	Value
Total wall time	221.2 minutes (3.69 h) — from log
Time per epoch	$\sim$33 seconds (derived)
GPU memory	not recorded
Best validation (sum of 5 MSEs)	$\mathbf{1.58 \times 10^{-4}}$ **at epoch 20**
Final validation	0.238 at epoch 400
Best checkpoint	`pretrained_v2.best.pt`
Final checkpoint	`pretrained_v2.pt`
Export (final weights)	`pretrained_v2.ibm.pt` (503 KB)

Best checkpoint (epoch 20) vs CFD: T and swirl



Best checkpoint (epoch 20) vs CFD: u_r, u_z



Best checkpoint in physical units (all 18,108 nodes)

Field	RMSE	Field std	R^2
u_r	0.22 mm/s	0.63 mm/s	0.875
u_θ (swirl)	1.58 mm/s	17.8 mm/s	0.992
u_z (axial)	0.18 mm/s	0.46 mm/s	0.841
p	24.2 Pa	1221 Pa	0.9996
T	0.47 K	17.7 K	0.9993

- Swirl, p and T are reproduced well; the two meridional velocities are the weak point (shared velocity scale).
- Maps cover all nodes, 95% of which were training points; the logged held-out numbers are on the previous slides.
- p is accurate to 24 Pa, but only 0.26 Pa of p is *not* hydrostatic (next sections): the network is mostly fitting a plane.

Rationale and What Was Actually Run

Pretraining used PennyLane's noiseless `default.qubit`. On real hardware the quantum features $\langle Z_i \rangle$ are perturbed by gate and readout errors, and the classical head was only ever trained on clean features.

Planned: freeze the circuit and fine-tune the classical head on hardware outputs.

Actually run: *inference only.* Before its training loop the script evaluates the model on 2,000 random points (seed 42) for a "predictions vs ground truth" plot; on a remote device each point is one hardware job. All 818 jobs were these evaluations.

Fine-tuning never started

`ibm_train.log` has no training lines, no fine-tuned checkpoint was written, and the plan time ran out first. Fine-tuning is **planned work**.

IBM Quantum Backend

Property	Value
Backend	**ibm_fez**, 156-qubit **Heron r2**
Calibration (24 Apr 2026, 18:54 CEST)	median CZ error 2.6×10^{-3}
	median SX error 2.8×10^{-4}
	median readout error 1.5×10^{-2}
Transpilation	optimisation level 1
Error mitigation	measurement twirling + measurement mitigation (TREX)
Dynamical decoupling	disabled
Shots per job	4,096 (precision 1/64)
Instance	hybrid_qcpinn (flex plan)

The Hardware Session

Metric	Value
Window	24 Apr 20:07 – 25 Apr 01:23 (UTC+2), 5.3 h
Jobs submitted	**818** (815 DONE, 3 cancelled)
QPU time (billed, DONE jobs)	9,826 s (2.7 h), $\approx$12 s per job
Mean submission rate	2.6 jobs/min; 46% of gaps $<$ 5 s
Distinct points evaluated	**303** of the 2,000 planned
Points run more than once	267 (up to 3 copies)

The repeats and the short gaps are consistent with two to three copies of the script running at once with the same seed (same points, same order). Plan time ran out at 01:23 (`ibm_train.log`). Earlier versions of this deck said 500 jobs because the job listing was capped at 500; all 818 were retrieved on 1 Oct 2026.

Transpiled Circuit on ibm_fez

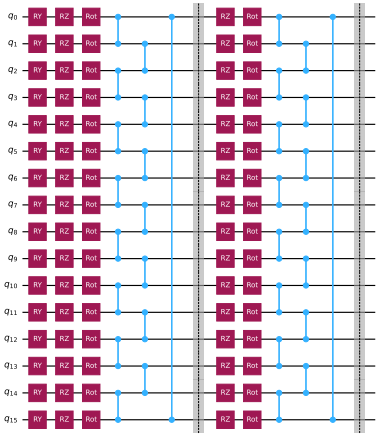
Logical circuit:

- 16 RY, 32 RZ, 32 Rot, **32 CZ**

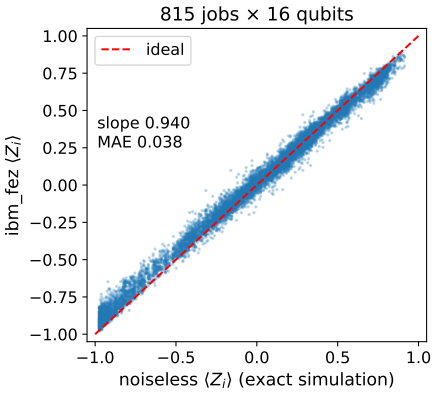
After transpilation (bound circuits, from the jobs):

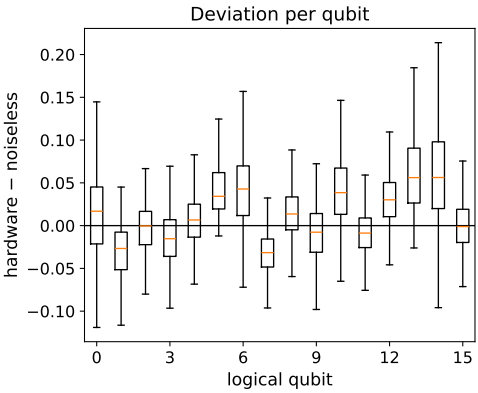
- CZ: 50–104 (median 83)
- SX: 96–196 (median 156)
- RZ: 96
- Depth: 30–90
- 16–23 physical qubits active (routing SWAPs)

This replaces the earlier "∼50 gates per shot" estimate.



Hardware vs noiseless $\langle Z_i \rangle$ (815 jobs $\times$ 16 qubits)





Hardware vs Noiseless: Numbers

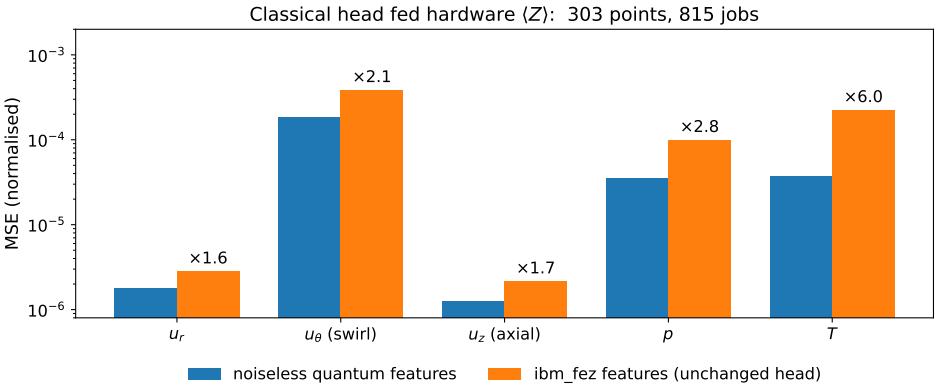
Noiseless reference: exact simulation of each job's own bound, transpiled circuit.

- Mean $|\text{hardware} - \text{noiseless}| = 0.038$ (RMSE 0.050), correlation 0.9966
- Hardware values are shrunk toward zero: fitted slope **0.94**
- Shot noise is only $\sigma \approx 0.014$; just 64% of deviations fall within 3σ — the rest is systematic hardware error
- Per-qubit bias reaches $+0.06$ (qubits 13, 14), -0.03 (qubits 1, 7)
- Repeats of the same point differ by 0.030 on average (std), about twice shot noise: run-to-run drift

Validation of the mapping

Every one of the 815 jobs matches `pretrained_v2.best.pt`'s noiseless features to $\sim 10^{-14}$; no other checkpoint matches. So all hardware jobs used the epoch-20 best checkpoint and the first 303 distinct points of the seed-42 sample.

Effect on predictions: classical head fed hardware features



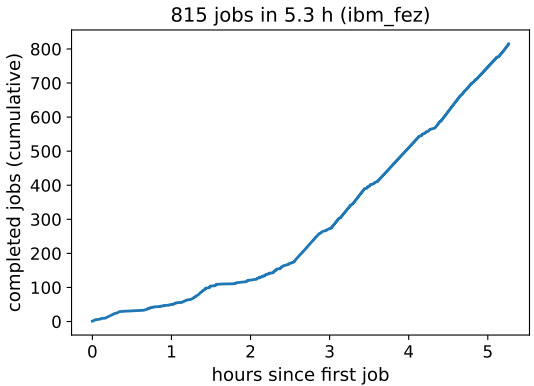
Effect on Predictions

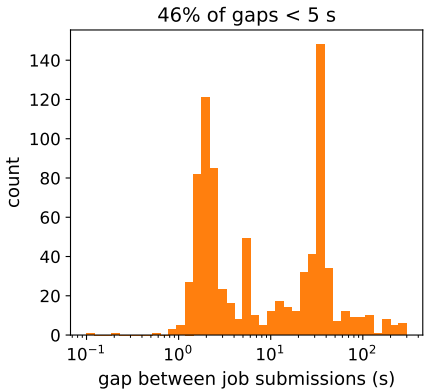
Classical head and weights unchanged (no fine-tuning); only the 16 quantum features replaced by hardware values.

RMSE	noiseless features	ibm_fez features
T	0.40 K	0.97 K
p	22 Pa	37 Pa
u_θ (swirl)	2.3 mm/s	3.3 mm/s
Mean of 5 normalised MSEs	5.2×10^{-5}	1.4×10^{-4}

- Hardware noise raises the mean MSE **2.7**×; T is hit hardest (6.0×).
- This is the one result the IBM time produced. 303 points, mostly training points; it measures noise sensitivity, not generalisation.
- Recovering this loss is what the planned head fine-tuning would target.

Job submission timeline (corrected)





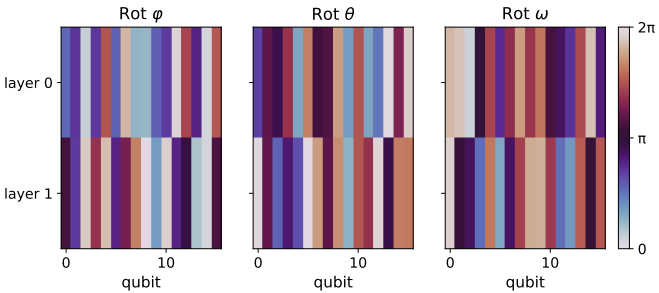
Status of the IBM Work

- **Done:** 815 hardware inference jobs, exact noiseless comparison, effect on predictions, real transpiled-circuit statistics, calibration snapshot.
- **Not done:** head fine-tuning on hardware (the planned Stage 3); no hardware-trained checkpoint exists.
- No QPU plan time remains. Job records and results were retrieved read-only on 1 Oct 2026 (the saved key still authenticated) and stored in `ibm_results/`.
- If hardware is available again: run the training loop *without* the 2,000-point diagnostic, and with one process only.

Quantum Circuit Weights (best checkpoint, epoch 20)

Statistic	Value
Shape	(2, 16, 3)
Mean	3.492 rad
Std	1.788 rad
Min	0.003 rad
Max	6.272 rad

Angles spread over $[0, 2\pi]$; none stuck at 0 or π. With $\sim$0.08% of the parameters this says little about whether they matter — the ablation says they do not.



Quantum LayerNorm (best checkpoint)

Statistic	Scale γ	Bias β
Min	0.938	-0.035
Max	1.002	$+0.023$
Mean	0.981	-0.0005

Interpretation

$\gamma \approx 1$ and $\beta \approx 0$: at epoch 20 the LayerNorm has barely moved from its initialisation, so all 16 qubit features are weighted nearly equally.

Classical Weight Distributions (best checkpoint)

Layer	Mean	Std	Max $\lvert w \rvert$
ff.B (Fourier, frozen)	−0.832	6.282	19.20
coord_proj.0	−0.001	0.083	0.263
coord_proj.2	0.001	0.057	0.186
res1.block.0 / .2	0.000 / 0.001	0.054 / 0.053	0.142 / 0.148
res2.block.0 / .2	0.000 / 0.000	0.053 / 0.052	0.138 / 0.174
to_quantum.0 / .2	−0.001 / 0.001	0.053 / 0.073	0.146 / 0.184
classical_skip.0	−0.001	0.052	0.133
post.0 / .2 / .4	0.000 / 0.001 / 0.002	0.065 / 0.053 / 0.067	0.229 / 0.141 / 0.140

All classical layers are zero-mean with small weights (std 0.05–0.08). Earlier versions of this slide mixed in values from the older v1 checkpoint (std 0.17–0.36).

Output Layer and Fourier Frequencies

Output layer (post.4) maps 64 hidden units $\to$ 5 channels:

- Mean $|w|$ per channel is 0.053–0.062: **no channel dominates**
- Output biases: -0.10, -0.05, $+0.03$, -0.11, -0.11 (the earlier "T bias $+0.22$" came from v1)

Fourier matrix B (frozen at initialisation):

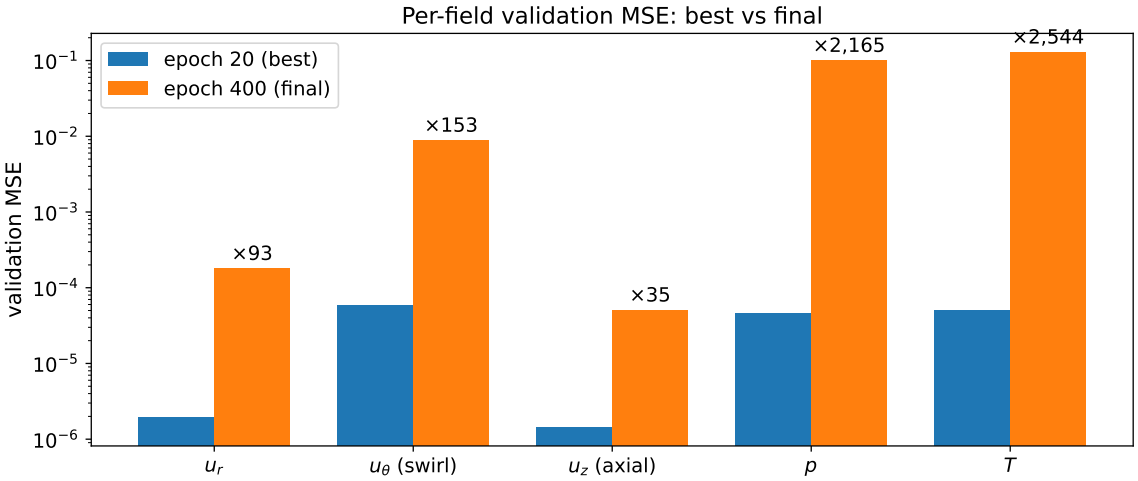
- Entries span about ± 19 (std 6.3, consistent with $\mathcal{N}(0, 6^2)$)
- The largest entry (19.2) is in the z row, column 0

Quantitative Impact

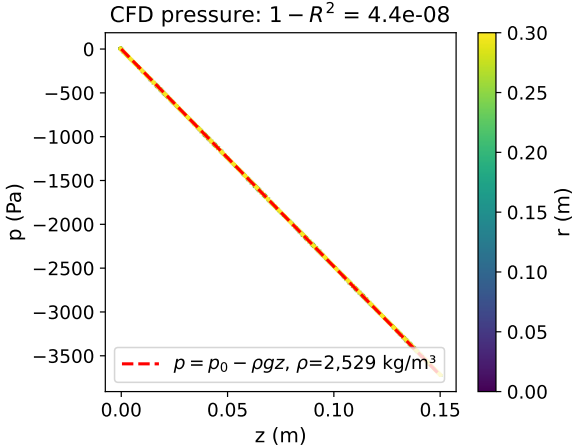
Field	Best (ep. 20)	Final (ep. 400)	Degradation
u_r	1.93×10^{-6}	1.79×10^{-4}	**93**×
u_θ (swirl)	5.78×10^{-5}	8.83×10^{-3}	**153**×
u_z (axial)	1.44×10^{-6}	5.06×10^{-5}	**35**×
p	$\mathbf{4.62 \times 10^{-5}}$	**0.100**	**2,165**×
T	$\mathbf{5.07 \times 10^{-5}}$	**0.129**	**2,544**×

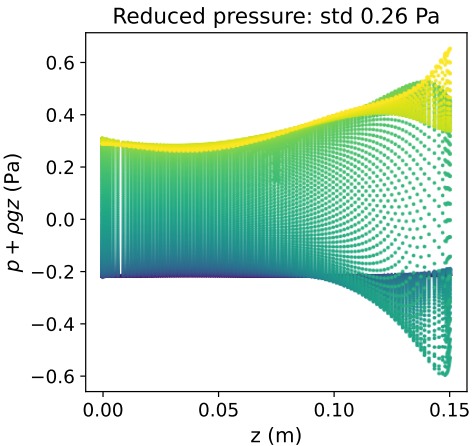
Normalised validation MSE. The variance of normalised p is 0.108 and of T is 0.074: the final p error (0.100) is that of a **flat prediction**, and the final T error (0.129) is **worse than predicting the mean**.

Per-field degradation, best vs final



The CFD pressure is purely hydrostatic





Root Cause: What the Physics Loss Asked For

- CFD p is hydrostatic: a linear fit in z gives slope $-24{,}811$ Pa/m, i.e. $\rho = 2{,}529$ kg/m^3 (liquid-silicon scale), $1 - R^2 = 4 \times 10^{-8}$.
- After removing it only **0.26 Pa** of variation remains, out of 3,722 Pa. "Predicting p" is fitting a plane.
- The implemented z-momentum has **no** ρg **term and** $\rho = 1$. For the CFD pressure the pressure-gradient term alone is $p_{coeff}\, \partial\hat{p}/\partial\hat{z} = -2.7 \times 10^5$, balanced by nothing. **The CFD solution itself violates this residual by** $\approx 2.7 \times 10^5$**.**
- Reducing that residual means flattening p; the network's final p is flat (MSE 0.100 vs variance 0.108).

Withdrawn

The earlier explanation ("p_{coeff} vs $1/Re$, a 2,000,000:1 imbalance") and its fix (weight p by $1/p_{coeff}^2$) are retracted: re-weighting cannot make an inconsistent equation consistent.

Defects in the v2 Run (`DEFECTS-2026-08.md`)

Found in August on MeLuXina CPU nodes; fixes were applied to the mainline scripts *after* the v2 checkpoint was trained.

1. **Transposed velocity columns.** The loader put the swirl column in the axial slot and vice versa. In the log, "u_z" is swirl and "u_θ" is axial; the physics residual was applied to the swapped fields too.
2. **One shared velocity scale**, which suppresses u_r and u_z by $64\times$ and $115\times$ in normalised space.
3. **EMA "balancing" is a no-op:** the physics term entered at 20 against a data loss of 2×10^{-5} — six orders of magnitude — and validation went from 2×10^{-4} to 0.24.

Reported in that document (not re-run here): with the fixes and magnitude-normalised balancing the physics-on fit reaches the data-only ceiling. Combined with the hydrostatic finding, this means the physics term then barely constrains p.

Loss Explosion at Epoch 280

```
[0280/400] loss=1.0368e+09 data=5.0592e-02 phys=2.1591e+10
[0290/400] loss=1.1795e+14 data=4.9662e-02 phys=2.1481e+15
[0300/400] loss=5.5708e-02 data=4.8543e-02 phys=9.6150e-02
```

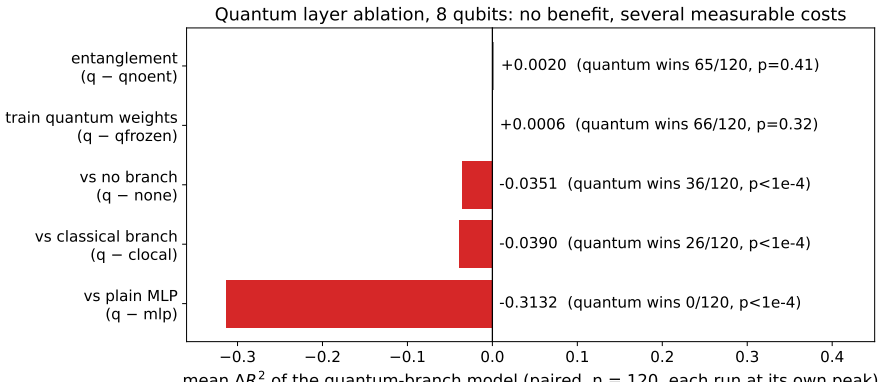
- Data loss stayed $\sim$0.05; the explosion is entirely in the physics residual
- Gradient clipping (max norm 1.0) bounded the update; recovered by epoch 300
- Cause not diagnosed (the earlier boundary-condition explanation was a guess)

Recommended Fix

Rebuild the physics loss before re-weighting it

1. Use the **reduced pressure** $p + \rho g z$ with $\rho = 2{,}529$ kg/m^3 and the real density in the momentum equations.
2. Check Re, Pr, Gr against the CFD case ($Pr = 28.463$ is far from liquid silicon, ~ 0.01).
3. Add boundary-condition terms.
4. Verify the residual on the CFD solution itself (it should be ≈ 0) *before* training.
5. Keep the loader fix, per-component velocity scales and magnitude-normalised balancing.
6. Report per-field R^2 in physical units, not a summed normalised MSE.

Does the quantum layer help? (8 qubits, 120 paired comparisons)



Ablation: Reading the Result

Separate harness on MeLuXina: supervised regression, sparse-data CZ task, quantum branch swapped for matched controls, each run at its own best epoch.

- Entanglement ($q - q_{\text{noent}}$): $+0.0020$, $p = 0.41$ — null
- Training the 48 quantum weights ($q - q_{\text{frozen}}$): $+0.0006$, $p = 0.32$ — null
- Worse than no branch (-0.035) and than a matched classical branch (-0.039), $p < 10^{-4}$
- A plain 21k-parameter MLP beats the 124k hybrid in **120 of 120** runs ($+0.31$ R^2)
- 12 and 16 qubits ($n = 12$ pairs): entanglement null at every size and budget, estimates drift negative with qubit count

Caveats: different harness, task and split from the H100 model here; 12/16-qubit rows exclude a large benefit but not a small one; one cell favours the hybrid (short budgets at high data density), the regime of this deck's epoch-20 checkpoint. Physics-on and the hardware stage were not tested.

Checkpoint Inventory

File	Epochs	Validation	Size	Use
`pretrained_v2.best.pt`	20 (of 400)	$\mathbf{1.58 \times 10^{-4}}$ (sum)	1.5 MB	**Best; used for all 815 IBM jobs**
`pretrained_v2.ibm.pt`	400	0.238	503 KB	Weights of the final epoch
`pretrained_v2.pt`	400	0.238	1.5 MB	Final checkpoint
`pretrained_16q.pt`	650, batch 128	not recorded	503 KB	v1
`fast_8q.pt`	120	not measured	495 KB	Mac pipeline check

v1 and the 8-qubit run have no validation metric: neither training script holds out data. All v2 checkpoints use the transposed-column loader; load them with that channel order.

HPC Resources (Brev H100 DGX)

Metric	Value
Total wall time	221.2 minutes (3.69 h)
GPU-hours	$\sim$3.7 H100 GPU-hours
Batches	68/epoch $\times$ 400 = 27,200
Training-sample circuit evaluations	17,203 $\times$ 400 $\approx$ 6.9M (data path)
GPU memory peak	not recorded
Cost	not recorded (3.7 GPU-h at the rate paid)
Best result	val sum $= 1.58 \times 10^{-4}$ (epoch 20)

IBM Quantum Hardware Resources

Metric	Value
Backend	ibm_fez (156-qubit Heron r2)
Jobs	818 (815 DONE, 3 cancelled)
Shots per job	4,096 nominal
QPU time billed	9,826 s (2.7 h)
Wall-clock window	5.3 h
Transpiled circuit	depth 30–90, CZ 50–104
Distinct points	303 (267 repeated)
Training completed	**none** (inference only)
Termination	flex plan time exhausted

Key Findings

1. **Best deployable model: the data-only epoch-20 checkpoint.** Swirl, p, T at $R^2 \geq 0.99$; u_r and u_z weaker (0.87, 0.84).
2. **The physics loss, as written, is inconsistent with the CFD data.** Pressure is hydrostatic ($\rho = 2{,}529$ kg/m^3), the loss has no ρg term, and the run also had three code defects.
3. **Real-hardware noise measured:** ibm_fez $\langle Z \rangle$ deviates by 0.038 on average (slope 0.94), raising prediction MSE 2.7$\times$ with an unchanged head.
4. **No evidence that the quantum layer helps** in the controlled ablation; a plain MLP wins 120/120.
5. **Hardware fine-tuning never ran**; the IBM time bought inference jobs only.

Next Steps

Priority	Action	Compute
Critical	Rebuild the physics loss (reduced p, real ρ, BCs); check residual on CFD	CPU
Critical	Retrain with the fixed loader and per-component scales	CPU/GPU
High	Classical-only baseline on the CZ data in this deck's setup	CPU
High	Hardware head fine-tuning, if QPU time is available	IBM plan
Medium	Parametric model (Re, Ω, v_{pull} inputs)	new CFD runs
Low	Deploy inference API	CPU only

Appendix A: Software Versions

Package	Version
Python	3.12 (Brev) / 3.14 (local)
PyTorch	2.11.0+cu130 (Brev training)
PennyLane	0.44.1 (training)
Qiskit / Qiskit-IBM-Runtime (April session)	2.3.0 / 0.45.0
Analysis (Oct 2026): Qiskit, runtime, PennyLane	2.5.2, 0.50.0, 0.45.1
numpy (Brev)	1.26.4

Appendix B: File Inventory

File	Description
`cz_melt_raw.txt`	CFD dataset (18,108 nodes)
`cg-hqpinn-brev.py`	H100 training script
`cg-hqpinn-ibm/CG_HQPINN_IBMtest_16q_effective.py`	IBM script (inference diagnostic ran; training did not)
`cg-hqpinn-mac-local.py`	8-qubit pipeline check
`make_figures*.py`, `make_circuit_fig.py`, `make_pred_cache.py`	Figures from checkpoint, log and CFD data
`ibm_analysis.py`, `ibm_map_points.py`, `ibm_plots.py`	Hardware vs noiseless, job-to-point mapping
`ibm_results/`	818 job records, bound circuits (QPY), analysis
`figures/`	Slide figures and `stats.json`
`pretrained_v2.best.pt` / `.pt` / `.ibm.pt`	v2 checkpoints
`brev_train.log`, `ibm_train.log`, `ibm_job_ids.json`	Logs; job IDs (first 500 only; all 818 in `ibm_results/`
`DEFECTS-2026-08.md`	Loader, scale and balancing defects

Appendix C: Unverified Items

- $Re = 15$, $Pr = 28.463$, $Gr = 8000$ as passed to the code: not checked against the COMSOL case set-up. Pr is far from liquid silicon.
- Mac pipeline check duration ($\sim$29 min): from notes, no log kept.
- H100 memory and cost: never measured.
- DEFECTS-2026-08.md results (fixed loader, physics on matching data-only): from MeLuXina logs, not re-run here.
- Cause of the epoch-280 loss spike.
- v1 (`pretrained_16q.pt`) hardware and accuracy: no log kept.

Appendix D: Corrections to the Earlier Version of This Deck

- Domain is 0.3×0.15 m, not 0.3×0.3.
- IBM: 818 jobs (not 500), 5.3 h (not 125 min), Heron r2 (not Eagle r3), measurement twirling/TREX (not M3); "62 forward passes", "2,048,000 shots", "$\sim$50 gates", "$\sim$100M gate operations" removed; fine-tuning did not run.
- Calibration-point protocol (8 points) never executed; its coverage figure is not shown.
- Physics ramp is 150 epochs (50$\rightarrow$200), not 200.
- Logged "val" is the sum of five MSEs; "u_z"/"u_θ" log labels are swapped.
- Root cause replaced (hydrostatic pressure, missing ρg, code defects); the 2,000,000:1 argument and $1/p_{coeff}^2$ fix withdrawn.
- Batches: 68/epoch (not 134). H100 memory and cost removed (not measured).
- Mac and v1 validation MSEs removed (no validation split); weight tables regenerated from the best checkpoint only.

Thank you

Questions?